



02 — Salience Dynamics

Overview

Salience models the brain's **memory strength** — a scalar that determines how likely a memory is to be sampled during replay, retrieved during recall, and survive decay. It is the product of three interacting forces: novelty at encoding, exponential recency decay, and recall-driven reinforcement.

Mathematical Formulation

State Variable

For each episodic memory i :

$$s_i(t) \in [s_{\min}, \infty), \quad s_{\min} = \text{min_salience}$$

Exponential Decay

Between time t_0 and t , salience decays exponentially with time constant τ :

$$s(t) = s(t_0) \cdot e^{-(t-t_0)/\tau}$$

$$\tau = \text{tau_seconds} \quad (\text{default: } 3600 \text{ s} = 1 \text{ hour})$$

Half-life: $t_{1/2} = \tau \cdot \ln(2) \approx 2495 \text{ s} \approx 41.6 \text{ min}$

Enforced floor:

$$s(t) = \max(s_{\min}, s(t))$$

Recall Reinforcement

Each time a memory is retrieved (and positively reinforced), salience increases additively:

$$s \leftarrow s + \Delta_r, \quad \Delta_r = \text{recall_reinforcement} \quad (\text{default: } 0.2)$$

This is applied per-episode during retrieval, only to the top-k cosine-direct episodes (not graph-harvested ones).

Consolidation Penalty

After a memory is consolidated into a prototype, its episodic salience is multiplicatively penalized:

$$s \leftarrow \max(s_{\min}, s \cdot \gamma_c), \quad \gamma_c = \text{consolidation_penalty} \quad (\text{default: } 0.5)$$

Brain analogue: consolidated memories in cortex no longer need strong hippocampal traces.

Proportional Sampling (for Replay)

During replay, episodes are sampled without replacement with probability proportional to salience:

$$P(i) = \frac{s_i}{\sum_j s_j}$$

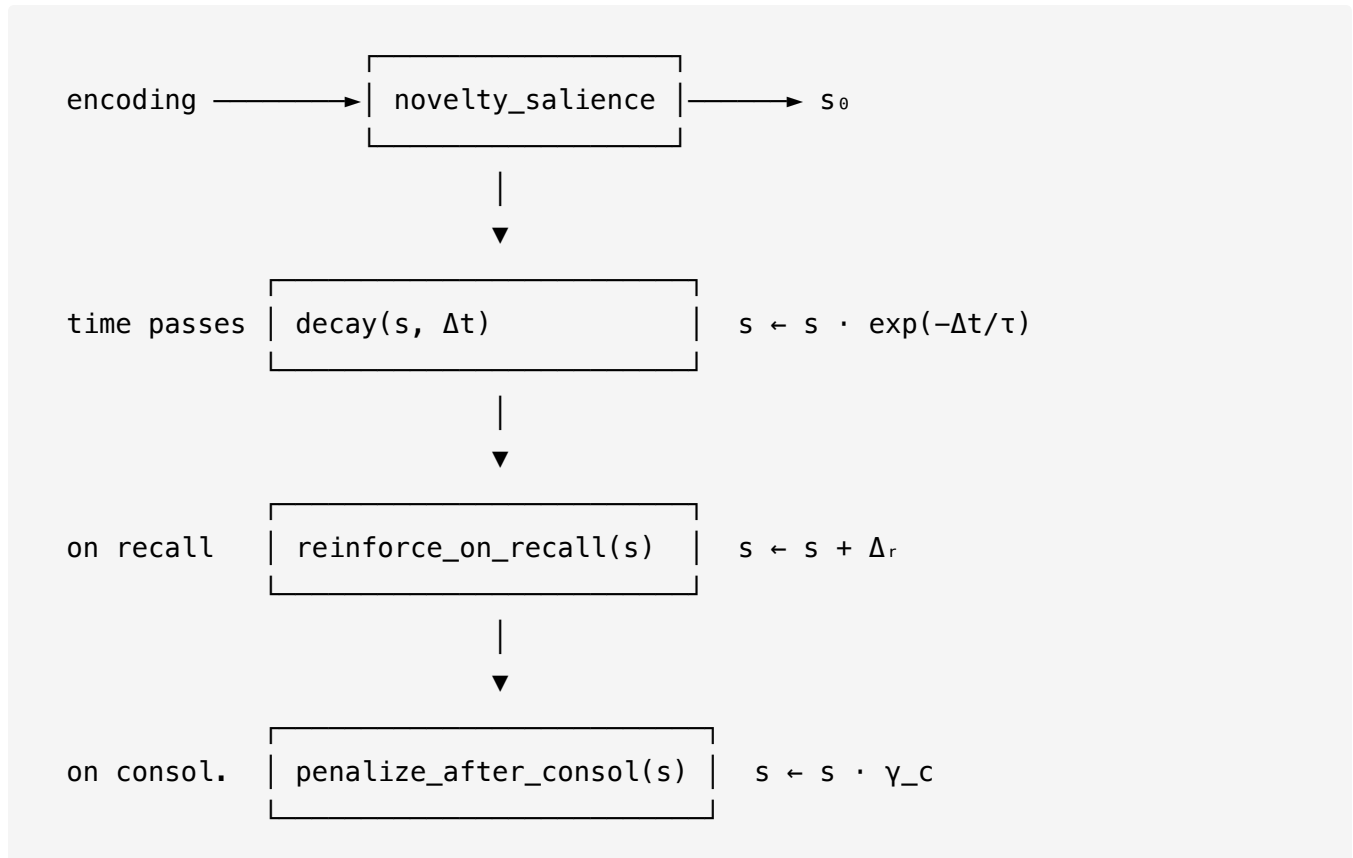
Where s_i are clamped to $\geq s_{\min}$ before normalization.

Configuration

SalienceConfig

Parameter	Type	Default	Description
tau_seconds	float	3600.0	Exponential decay time constant (seconds)
min_salience	float	0.01	Absolute floor for salience
novelty_weight	float	1.0	Weight for novelty during initialization
recall_reinforcement	float	0.2	Additive boost per recall event
consolidation_penalty	float	0.5	Multiplicative penalty after consolidation

Computational Pipeline



Key Invariants

1. Saliency never drops below `min_salience` — no memory is ever fully erased.
2. Decay is purely exponential; reinforcement is purely additive — they compose sequentially.
3. Recall reinforcement is only applied to cosine-direct episodes, preventing graph-harvested self-reinforcement loops.
4. Consolidation penalty ensures the hippocampus cedes control to cortical prototypes.